TIDES Phase 5

Multi-Region Coordination & Intelligent Rescue Resource Allocation

Implementation brief | 22 October 2026 | Decision-support simulation only

Objective

Connect the existing multi-beach registry, regional incidents, forecast context, rescue resources, drones, and command center through one explainable, read-only operational coordination layer.

Reused Architecture

BeachRegistry remains the authoritative multi-region site registry.

RegionalIncidentAdapter remains the incident source and preserves site identity.

RegionalResourceRegistry remains the resource source and exposes availability/provenance.

Existing mutual-aid and dispatch services remain responsible for operator-approved actions.

Existing wristband, distress, orange locator, drone, and rescue state machines are unchanged.

Phase 5 Integration

Cross-region overview

Phase5CoordinationService aggregates registered sites, active incidents, unassigned incidents, resource availability, and stale/disconnected state without averaging away local hazards.

Incident prioritization

Active incidents are ordered deterministically using established severity, distress level, elapsed age, and incident ID tie-breaking. Each result includes an explanation. Forecast context is explicitly separate from individual distress.

Resource recommendations

Recommendations include only available resources with fresh registry data. Station-bound resources are excluded from cross-beach use. Cross-beach candidates remain advisory and require operator authorization and transit review.

Command center

The existing regional command center now displays a Phase 5 operations strip with active priority incidents, site identifiers, eligible resource count, and the advisory-only dispatch boundary.

Files

Verification

Known Limitations

Resource positions, capabilities, and site records are simulated or demonstration data where marked by the registry.

No autonomous cross-region dispatch is performed.

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